An AI-Ready Air-to-Ground Channel Model for Multi-Link HAPS–UAV Networks

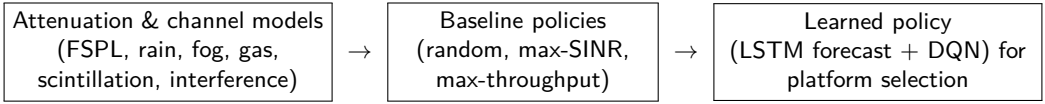
Harsha Matadhikari Prof. Sonali Garg Prof. Abhishek Dixit

Online PG Diploma in Advanced Communication Engineering with Quantum AI Integration
Indian Institute of Technology Delhi

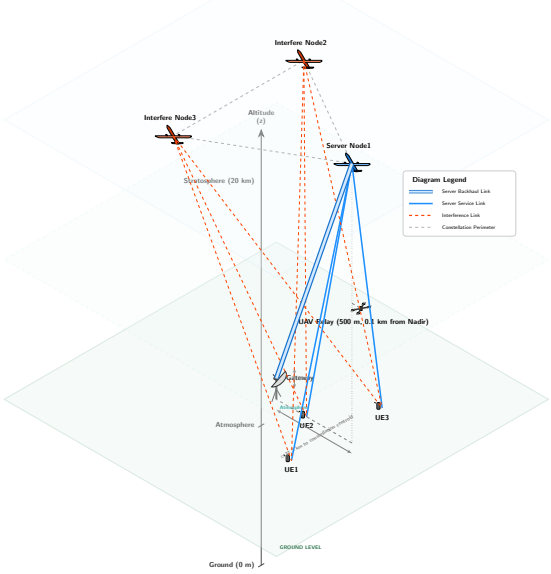
Motivation

- A 3-HAPS equilateral-triangle constellation serves a ground footprint via **service** (2 GHz), **backhaul** (38 GHz), and two-hop **UAV relay** links.
- Co-channel interference from the other two HAPS makes SINR depend strongly on *angular position*, not just distance – an experimental finding of this paper, not prior knowledge.
- A simple rule-based policy cannot anticipate this, time-varying rain fades on the 38 GHz backhaul, or battery depletion.
- **AI Ready Model:** a validated channel model, an LSTM rain forecaster, and a DQN agent that learns real-time user–platform association.

Research flow:



System Topology and Channel Model



Three link types:

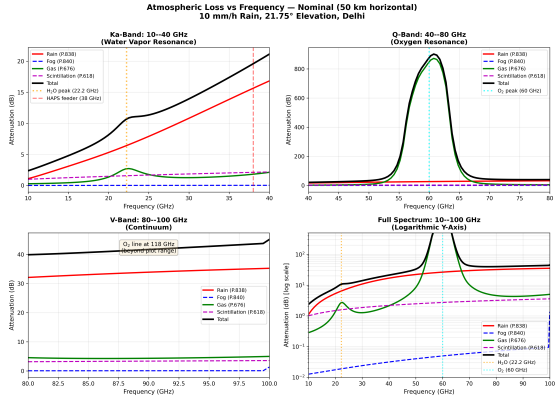
- HAPS → UE (service, 2 GHz)
- HAPS → Gateway (backhaul, 38 GHz)
- HAPS → UAV → UE (relay; feeder hop at 38 GHz, UAV→UE access hop at 2 GHz)

Channel model:

- Free-space path loss (all links)
- Building-blockage LoS/NLoS probability (2 GHz UAV/HAPS access tier, Holis & Pechac 2008)
- ITU-R rain, fog, gas, and scintillation losses (38 GHz backhaul/feeder)

Topology: 3 HAPS (20 km, 65 km equilateral triangle), 1 static UAV relay (500 m AGL, fixed 0.1 km from the serving HAPS's own nadir), 1 gateway (50 m AGL, fixed at the triangle's centroid instead)

Why 38 GHz: Frequency Selection



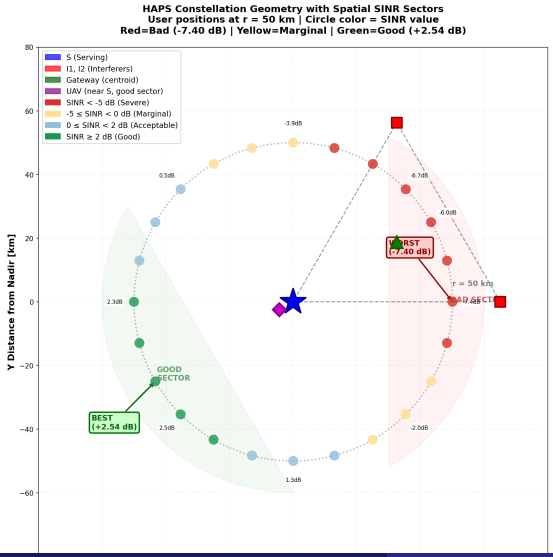
Atmospheric loss (rain, fog, gas) swept across the Ka/Q/V bands for the Delhi gateway path, 10 mm/h rain:

- **Rain** (P.838/P.618) – dominant term, grows roughly linearly with frequency; up to ≈19 dB at 99.9% availability ($R_{0.1} = 15.7$ mm/h).
- **Fog/cloud** (P.840) – small except in dense fog; negligible below 10 GHz.
- **Gas absorption** (P.676) – two resonance peaks to avoid: H_2O at 22.2 GHz, O_2 at 60 GHz (there, total loss exceeds 800 dB over 50 km – unusable).

38 GHz sits in the valley between these peaks (≈0.2–0.3 dB/km clear-sky gas loss) – this, not just available bandwidth, is why it was chosen over a frequency closer to either resonance.

Backhaul budget: ≈38 dB clear-sky SINR comfortably covers ≈19.4 dB of rain+gas loss, leaving ≈18.6 dB

Angular Interference Sectors and Scintillation Fading



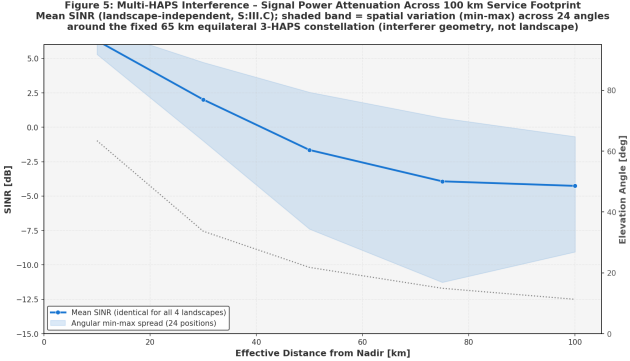
Angle-dependent SINR (sectors): the 3-HAPS triangle creates a good sector (≈120°–290°, away from interferers) and a bad sector (≈330°–75°, toward them) – same radius, up to 9.94 dB apart.

Scintillation (ITU-R P.618): rapid amplitude fluctuations from tropospheric turbulence; grows with frequency, shrinks with elevation. At 30° elevation, $p = 0.01\%$: ≈0.49 dB at 38 GHz vs. ≈0.09 dB at 2.1 GHz.

Favors positioning the UAV relay near nadir (higher elevation) when feasible, especially for the HAPS–UAV feeder hop – which also tends to be away from the interferer-facing sector.

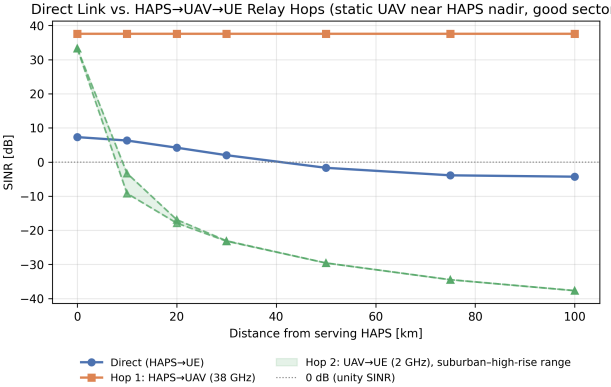
Nuance: each HAPS could simultaneously be a *serving* node (near-nadir, low-loss for its own users) and an *interferer* for the other two HAPS' users – so this isn't a single clean optimization across the constellation (future work, see Conclusion).

Spatial SINR Heterogeneity



- At a **fixed 50 km radius**, SINR varies by **up to 9.94 dB** depending on angular position relative to the two interfering HAPS.
- For comparison, simply *doubling* the link distance only costs about 2–3 dB – so angular position affects SINR more than distance does.
- A simple "always pick max-SINR" rule cannot anticipate these angular patterns as a user moves or the constellation's geometry is reconsidered.
- Backhaul link stays noise-limited and essentially flat; the service link is the one exposed to this heterogeneity.

Relay Path Analysis (HAPS→UAV→UE)



- Single static UAV, fixed 0.1 km from the serving HAPS's own nadir (the Gateway occupies the centroid instead).
- **Feeder hop** (HAPS→UAV, 38 GHz): constant geometry by symmetry – flat at 12.48 bits/s/Hz, ≈89.71° elevation, for every HAPS.
- **Access hop** (UAV→UE, 2 GHz): collapses within a few km of the fixed UAV – far faster than the direct link degrades with distance from a HAPS.
- Raw capacity: relay exceeds the direct link within ≈1–6 km of the serving HAPS's own nadir, falling to 0.50 bits/s/Hz (direct: ≈2.5–2.6) at R_{eff}.
- Contention: one HAPS's direct-link population ($100^2/3$, full footprint)

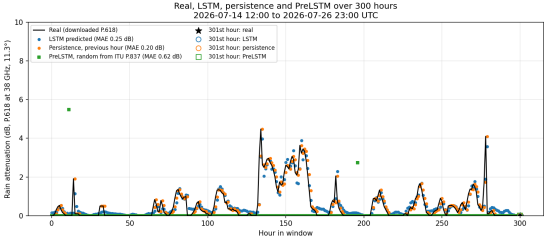
Motivation for Learning

A rule-based policy cannot account for three sources of variation this paper's channel and platform models expose:

- **Spatial SINR heterogeneity.** Angular position affects SINR more than distance – a simple "always pick max-SINR" rule cannot anticipate it as a user moves.
- **Time-varying weather.** Feeder-link rain attenuation must be anticipated before the hour it affects, since the association decision is made before that hour's rain is observed.
- **Battery depletion.** Each HAPS's state of charge depends on its load history, so load must be balanced across platforms over time to avoid over-draining any one HAPS.

A learned, real-time association policy can condition on all three jointly.

LSTM Rain-Attenuation Forecaster



- DQN must decide at hour t *before* that hour's rain is observed $\Rightarrow$ needs a forecast.
- Early-stopped on held-out 2025 validation RMSE to avoid overfitting; 2026 untouched until final evaluation.
- Evaluated **out-of-sample**: 2026 forecasts from a model trained only on 2023–2025 data.

- LSTM predicts feeder (38 GHz) rain attenuation up to **six hours ahead** from NASA POWER / METAR / ERA5 weather history for Delhi.
- Lower RMSE than persistence at every horizon except +1 h, where persistence is slightly better; all six horizons feed the DQN's state.

Horizon	LSTM RMSE	Persist. RMSE
+1 h	0.54	**0.53**
+3 h	**0.80**	0.90
+6 h	**0.92**	1.17

DQN Association Agent: Problem Formulation

Per-user MDP – each user independently chooses a platform; network reward is computed after all assignments. With a single static UAV, which HAPS feeds its feeder hop is folded into the same per-user action set.

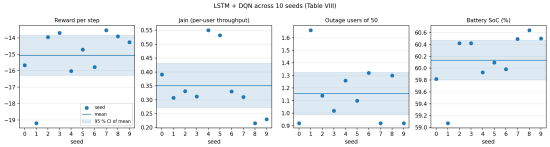
- **State (21-dim):** SINR to each platform, distance, battery SoC, UAV-relay capacity, rainfall, visibility, time-of-day, six LSTM rain forecasts.
- **Action:** $a_k \in \{0, \ldots, 5\}$ – direct link to HAPS1–3, or relay through the UAV served by HAPS1–3.
- **Reward:** rain-derated throughput + fairness (Jain index) − energy cost − outage/degraded-service penalties − low-battery penalty.
- Custom Deep Q-Network implemented directly in PyTorch (`torch/torch.nn`).
- Trained on 2400 rolling 24-hour episodes (400 real 2024 episodes, cycled six times); evaluated on 50 held-out 2025 days across ten experiments (random seeds).

Results: Policy Comparison

Policy	Reward/step	Jain	Outage /50
Random (six actions)	−92.2 (1.4)	0.255 (0.011)	41.6 (0.3)
Max-SINR (direct only)	−14.5 (1.7)	**0.558** (0.013)	**0.8** (0.2)
Max-throughput (six actions)	**−14.6** (1.7)	0.241 (0.039)	**0.8** (0.2)
LSTM + DQN	−15.1 (1.7)	0.351 (0.112)	1.2 (0.2)

- Mean ± std. dev. over ten experiments, 50 held-out 2025 days.
- LSTM+DQN is behind both rule-based baselines on reward and outage, and behind max-SINR on fairness, but **fairer than max-throughput** (Jain 0.351 vs. 0.241, 8/10 seeds).
- Max-SINR and max-throughput both need continuous, perfect, zero-latency measurement of every platform's SINR or capacity from every UE – an idealized assumption, not something a real UE can actually provide. Loading all crowded users onto max-throughput's single best option is also why its own fairness collapses (0.241) once contention is modelled.
- The DQN uses only its own locally available state and allows dynamic switching of links based on practical, learned-from-past-data behaviour – the only one of the three that is actually deployable, and the only one that retains and uses the relay path.

Results: Stability Across Experiments



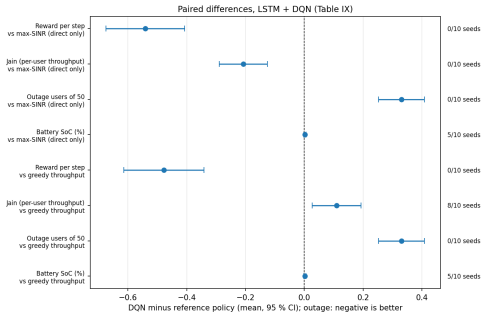
LSTM + DQN, ten independent seeds (dot = one training run; line = mean; band = 95% CI *of the mean*):

- Reward/step: -15.07 ± 1.71 (95% CI $[-16.30, -13.85]$)
- Jain fairness: 0.351 ± 0.112
- Outage: 1.16 ± 0.24 of 50 users
- Battery SoC: 60.14 ± 0.47 %

CI vs. spread: the shaded band is *not* "most runs land here" – it narrows with more seeds and reflects precision of the estimated mean, not run-to-run spread.

Takeaway: the tight spread across all ten seeds shows the policy is **reliably learned** to this level, not a lucky or unlucky single run – consistency, not the reward level itself, is what this figure demonstrates.

Results: Statistical Significance



Paired differences, LSTM+DQN minus baseline (per experiment, 50 days each, ten seeds). Whiskers crossing 0 = not significant.

vs. Max-SINR:

- Reward: -0.54 ($p < 0.0001$, 0/10 seeds favoring DQN)
- Jain: -0.207 ($p = 0.0003$, 0/10)
- Outage: $+0.33$, i.e. worse ($p < 0.0001$, 0/10)
- Battery SoC: ≈ 0 ($p = 0.33$, tie)

vs. Max-throughput:

- Reward: -0.48 ($p < 0.0001$, 0/10)
- Jain: $+\mathbf{0.110}$, i.e. **fairer** ($p = 0.0153$, 8/10)
- Outage: $+0.33$, i.e. worse ($p < 0.0001$, 0/10)
- Battery SoC: ≈ 0 ($p = 0.33$, tie)

Takeaway: the DQN loses to both rule-based policies on reward and outage on every seed, and to max-SINR on fairness on every seed, but is **fairer than max-throughput** on 8/10 seeds – and still uniquely uses the relay path as a targeted fallback.

Does the Rain Forecast Help the DQN Itself?

A forecast ablation (`lstm` vs. `none`, rain-forecast features removed, vs. `oracle`, true future attenuation given) finds all three statistically indistinguishable on reward, fairness, and outage ($p > 0.1$) – and it stays null after two further tests:

- **Gave the UAV feeder its own rain fade.** Previously unmodelled; its near-zenith path actually fades *more* than the backhaul's at most rain rates. Still null.
- **Added a per-user switching-cost penalty.** Tests whether reacting late (vs. anticipating) is the missing friction. Still null – with one UAV, there is no penalty-free alternative to pre-emptively move traffic to.
- **Why:** this MDP lets every user react to the current hour's rain for free – raw rain rate is already a state feature regardless of the forecast. A forecast only pays off when reacting late costs something real.
- **Where it would matter:** a fleet, where a fade over one HAPS squeezes every UAV feeding off it at once – forcing a synchronized surge of handovers unless the policy can see it coming and stagger them.

Takeaway: rain does not impact runs on this geometry – the 2 GHz access link and the margin-supported 38 GHz feeder link both absorb it without needing to anticipate it. The forecaster's behaviour against persistence is nonetheless a useful input for more complex geometries where margins no longer absorb rain's losses.

Conclusion and Future Work

Backhaul. Noise-limited, largely unaffected by constellation geometry, and largely insulated from rain by its design margin. **Service.** Spatial SINR heterogeneity (up to 9.94 dB at fixed 50 km) dominates quality; cannot multiplex simultaneous users, so cannot scale to replace the relay. **Relay.** Access hop collapses within a few km; but once contention is modelled (one HAPS's direct-link crowd is ≈79x a UAV's relay crowd, by geometry alone), relay beats direct *everywhere* inside its own disk. One static UAV still only covers that one disk (≈6.5 km radius; ∼200–270 UAVs for full coverage). **Learned association.** Measurably, if narrowly, behind both rule-based baselines on reward ($p < 0.0001$), but using only its own locally available state, not the continuous, perfect, zero-latency global measurement the baselines require – the only one of the three that is actually deployable.

Roadmap for future work:

- *Model multiple UAVs:* positioning for best SINR, UAV→HAPS association for N UAVs/M HAPS, and a UAV-to-UAV interference term plus relative motion of HAPS and UAVs.
- *LSTM and DQN improvements:* rain forecasting may help once a fleet of UAVs changes the action space; other impairments (fog, scintillation, gas) are not yet analysed as time series; the DQN's feeder-HAPS choice should condition on a per-UAV, physically grounded feeder-channel state rather than the single global mean user state used here.
- *Dynamic topology and geometry:* changing HAPS roles (serving vs. interfering) and jointly optimizing positioning across both; beamforming between HAPS and gateway.

Thank you – Questions?